

Report

- Assignment 3 -

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Dataset Description

Fashion-MNIST is a dataset consisting of Zalando's article images, intended as a more challenging replacement for the original MNIST dataset. It includes a training set of 60,000 examples and a test set of 10,000 examples. Each example is a 28×28 grayscale image, associated with a label from one of 10 clothing categories.

Each pixel in the image has a single integer value between 0 and 255, indicating its brightness — where higher values represent darker pixels.

The data is often provided in CSV format, where each row corresponds to one image:

- The first column contains the class label, indicating the clothing category.
- The remaining 784 columns represent the flattened pixel values ($28 \times 28 = 784$).

The class labels are as follows:

- 0: T-shirt/top
- 1: Trouser
- 2: Pullover
- 3: Dress
- 4: Coat
- 5: Sandal
- 6: Shirt
- 7: Sneaker
- 8: Bag
- 9: Ankle boot

Method for Dataset Division

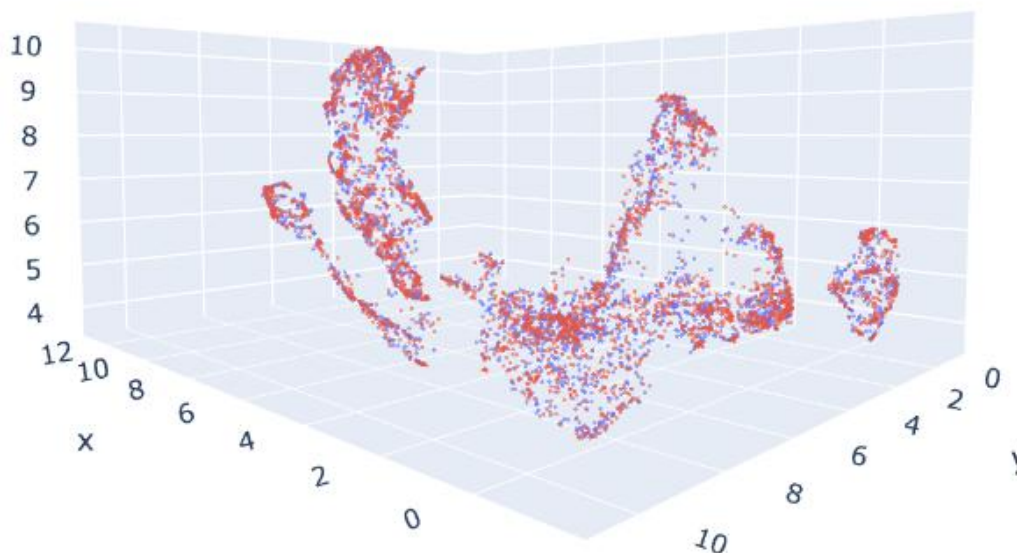
I believe that distribution shift in the dataset is a critical factor when training machine learning models. Since a model learns based on the distribution of the training data, a mismatch between the distributions of the training and testing datasets is likely to degrade performance. Therefore, I followed the train/test split provided by Kaggle and investigated whether there is a distribution shift between them. Specifically, I compared the distributions of 'fashion-mnist_train.csv' and 'fashion-mnist_test.csv'.

Methods for Detecting Data Distribution Shifts

To gain an intuitive understanding of potential distribution differences, I visualized the merged train and test datasets using UMAP. However, since visualizations alone may not clearly reveal distributional discrepancies, I also conducted a quantitative comparison using the Classifier Two-Sample Test (C2ST). For simplicity, logistic regression was used as the classifier, and accuracy was employed as the evaluation metric.

Results & Interpretation

First, the result of the UMAP visualization is shown below. Blue points represent the training data, and red points represent the test data.



The distinct clusters—regardless of the point color—are presumed to correspond to different classes. Based on the visualization alone, there does not appear to be a significant distribution shift between the two datasets. Therefore, to quantitatively assess the difference, I performed a Classifier Two-Sample Test (C2ST). Using logistic regression

as the classifier, the test yielded an accuracy of 0.8579.

Through the results of C2ST, it can be confirmed that there exists a distribution shift between the train and test datasets, although this shift is not clearly visible through visualization. I believe this is because UMAP reduces the dimensionality of the data, which causes the information regarding the distribution differences between the two datasets to be diminished.

Furthermore, I consider the type of distribution shift between these datasets to be a covariate shift. The distribution differences arise from variations in the appearance of each piece of clothing, i.e., the covariates. Since the labels remain unchanged despite changes in the covariates, it is reasonable to classify this as a covariate shift.

Due to this distribution difference, when the validation set is created from the train set and a deep learning model is trained accordingly, the performance metric on the test set may be lower than that on the validation set.

Discussion on Monitoring & Mitigation Methods

There are various methods for monitoring distribution shifts. The visualization technique I used in this case, such as UMAP, and statistical tests like C2ST can also be employed for monitoring. In addition, feature-wise distribution comparison, which involves visualizing the distribution of each feature using histograms or density plots, can be used. Another approach is to utilize drift detection libraries that are designed to detect shifts in data distributions.

Once a distribution shift is detected through these methods, several techniques can be used for mitigation. A simple approach is to perform data rebalancing by combining all datasets, shuffling them, and then re-splitting. In other cases, techniques such as data augmentation can be applied to increase the size and diversity of the dataset, thereby helping to reduce the distribution gap.

Link for the remote source repository

<https://github.com/twoSilverUp/MLOps/tree/Assignment3>